Powers of Ten

Times 10: every digit moves one place to the left.

34 × 10 = 340

thousands	hundreds	tens	ones		tenths	hundredths	thousandths
		3	4	.			
	3	4	0	.			

Each digit is worth 10 times as much, so it moves one place left.

The empty ones place gets a 0.

4.6 ÷ 10 = 0.46

			4	. 6		
			0	. 4	6	

Divide by 10: one place right.

Divide by 100: two places right.

Powers of ten: how many 10s you multiply

$10^2 = 10 \times 10 = 100$

$10^3 = 10 \times 10 \times 10 = 1{,}000$

$10^4 = 10{,}000$

The small number says how many 10s, and how many zeros.

Your turn!

$$5.2 \times 100 = \square$$

Powers of Ten: Practice

Times: digits move left. Divide: digits move right.

Part 1 Move the digits Write the new number in the second row.

1)

th	h	t	o	tth	hth	thth
		2	7			

27 × 10 = ☐

2)

th	h	t	o	tth	hth	thth
			6	4		

6.4 ÷ 10 = ☐

3)

th	h	t	o	tth	hth	thth
			3	1	5	

3.15 × 100 = ☐

4)

th	h	t	o	tth	hth	thth
	8	2	0			

820 ÷ 100 = ☐

5)

th	h	t	o	tth	hth	thth
			0	9		

0.9 × 10 = ☐

Powers of Ten: Practice (continued)

Part 1 Move the digits (continued)

6)

th	h	t	o	.	tth	hth	thth
	4	0	5	.			
				.			

405 ÷ 10 = ☐

Part 2 Times 10, 100 and 1,000 No help this time.

7) **47 × 100 =** ☐

8) **3.6 × 10 =** ☐

9) **0.25 × 100 =** ☐

10) **8 × 1,000 =** ☐

11) **5.04 × 100 =** ☐

12) **62 × 10 =** ☐

13) **0.7 × 1,000 =** ☐

14) **19 × 1,000 =** ☐

15) **2.38 × 10 =** ☐

Powers of Ten: Practice (continued)

Part 2 Times 10, 100 and 1,000 (continued)

16) $0.06 \times 100 =$ ☐

17) $715 \times 10 =$ ☐

18) $4.5 \times 1{,}000 =$ ☐

19) $0.81 \times 10 =$ ☐

20) $33 \times 100 =$ ☐

21) $1.9 \times 100 =$ ☐

Part 3 Divide by 10, 100 and 1,000 No help this time.

22) $4{,}500 \div 100 =$ ☐

23) $36 \div 10 =$ ☐

24) $7 \div 100 =$ ☐

25) $8{,}000 \div 1{,}000 =$ ☐

26) $52 \div 100 =$ ☐

27) $9.4 \div 10 =$ ☐

Powers of Ten: Practice (continued)

Part 3 Divide by 10, 100 and 1,000 (continued)

28) $630 \div 10 =$ ☐

29) $1{,}250 \div 1{,}000 =$ ☐

30) $3 \div 10 =$ ☐

31) $74 \div 100 =$ ☐

32) $580 \div 100 =$ ☐

33) $2.6 \div 10 =$ ☐

34) $9{,}000 \div 100 =$ ☐

35) $45 \div 1{,}000 =$ ☐

36) $0.8 \div 10 =$ ☐

Part 4 Powers of ten No help this time.

37) $10^2 =$ ☐

38) $10^4 =$ ☐

Powers of Ten: Practice (continued)

Part 4 Powers of ten (continued)

39) $10^1 =$ ☐

40) $10^5 =$ ☐

41) $3 \times 10^2 =$ ☐

42) $7 \times 10^3 =$ ☐

43) $2.5 \times 10^2 =$ ☐

44) $10^3 =$ ☐

45) $6 \times 10^4 =$ ☐

46) $4.2 \times 10^1 =$ ☐

Part 5 Times what? No help this time. Count how many places the digits moved.

47) $52 \times$ ☐ $= 5{,}200$

48) $3.7 \times$ ☐ $= 37$

49) $0.4 \times$ ☐ $= 400$

Powers of Ten: Practice (continued)

Part 5 Times what? (continued)

50) $16 \times \square = 16{,}000$

51) $2.09 \times \square = 209$

52) $81 \times \square = 810$

53) $0.35 \times \square = 35$

54) $7 \times \square = 700$

Part 6 Times 10, 100 and 1,000

55) $682 \times 1{,}000 = \square$

56) $607 \times 100 = \square$

57) $237 \times 10 = \square$

58) $310 \times 100 = \square$

59) $55.8 \times 100 = \square$

60) $34.6 \times 100 = \square$

Powers of Ten: Practice (continued)

Part 6 Times 10, 100 and 1,000 (continued)

61) $5.45 \times 10 =$ ☐

62) $2.15 \times 1{,}000 =$ ☐

63) $4.16 \times 1{,}000 =$ ☐

64) $52 \times 10 =$ ☐

65) $539 \times 1{,}000 =$ ☐

66) $4.96 \times 1{,}000 =$ ☐

67) $17.2 \times 10 =$ ☐

68) $3.1 \times 1{,}000 =$ ☐

69) $985 \times 100 =$ ☐

70) $264 \times 1{,}000 =$ ☐

71) $385 \times 10 =$ ☐

72) $8.32 \times 100 =$ ☐

Powers of Ten: Practice (continued)

Part 6 Times 10, 100 and 1,000 (continued)

73) $405 \times 10 =$ ☐

74) $3.16 \times 100 =$ ☐

75) $1.35 \times 1{,}000 =$ ☐

76) $349 \times 1{,}000 =$ ☐

77) $3.32 \times 10 =$ ☐

78) $58 \times 100 =$ ☐

Part 7 Divide by 10, 100 and 1,000

79) $6{,}898 \div 100 =$ ☐

80) $245 \div 100 =$ ☐

81) $3{,}505 \div 1{,}000 =$ ☐

82) $7{,}911 \div 1{,}000 =$ ☐

83) $5{,}342 \div 10 =$ ☐

84) $3{,}504 \div 100 =$ ☐

Powers of Ten: Practice (continued)

Part 7 Divide by 10, 100 and 1,000 (continued)

85) 7,984 ÷ 1,000 = ☐

86) 9,978 ÷ 10 = ☐

87) 6,099 ÷ 10 = ☐

88) 7,192 ÷ 1,000 = ☐

89) 6,087 ÷ 1,000 = ☐

90) 5,447 ÷ 1,000 = ☐

91) 3,345 ÷ 10 = ☐

92) 5,160 ÷ 10 = ☐

93) 1,694 ÷ 100 = ☐

94) 9,459 ÷ 1,000 = ☐

95) 3,635 ÷ 10 = ☐

96) 9,926 ÷ 1,000 = ☐

Powers of Ten: Practice (continued)

Part 7 Divide by 10, 100 and 1,000 (continued)

97) $1{,}450 \div 100 = \square$

98) $9{,}342 \div 10 = \square$

99) $3{,}067 \div 100 = \square$

100) $9{,}635 \div 1{,}000 = \square$

101) $5{,}628 \div 10 = \square$

102) $1{,}958 \div 100 = \square$

Part 8 Remember? Problems from earlier packets.

103) $0.62 \; \square \; 0.6$

104) $4 \times \frac{7}{10} = \frac{\square}{10}$

105) area: $18 \times 7 = \square$ sq ft

106) $48 \times 25 = \square$

107) $6 \text{ L} = \square$ mL

108) $40 + \square = 180$ degrees

Powers of Ten: Practice (continued)

Part 9 Word problems Think about how the digits move.

109) A box holds 24 pencils. How many pencils are in 100 boxes?

Number sentence _____

Answer ☐ pencils

Draw or show your work

110) A stack of 10 coins is 1.8 cm tall. How tall is one coin?

Number sentence _____

Answer ☐ cm

Draw or show your work

111) A sticker costs $0.35. How much do 10 stickers cost?

Number sentence _____

Answer ☐

Draw or show your work

112) A farm has 4,500 apples to pack in 100 equal bags. How many apples go in each bag?

Number sentence _____

Answer ☐ apples

Draw or show your work

113) Challenge: A paper is 0.1 mm thick. How thick is a stack of 1,000 papers, in cm? (10 mm = 1 cm)

Number sentence _____

Answer ☐ cm

Draw or show your work

Check-In: Powers of Ten

Show what you know. No help this time. Score: _____ / 9

Part 1 Move the digits

1)

th	h	t	o		tth	hth	thth
			5	.	2		
				.			

$5.2 \div 100 =$ ☐

Part 2 Times 10, 100 and 1,000

2) $13 \times 10 =$ ☐

3) $9.1 \times 1{,}000 =$ ☐

4) $740 \times 10 =$ ☐

Part 3 Divide by 10, 100 and 1,000

5) $56 \div 10 =$ ☐

6) $4{,}100 \div 100 =$ ☐

7) $4{,}100 \div 10 =$ ☐

Part 4 Powers of ten

8) $9 \times 10^3 =$ ☐

9) $1.5 \times 10^3 =$ ☐